[NOT MEASUREMENT SENSITIVE]

P-W-2891
31 JULY 1996

FEDERAL SPECIFICATION

WIPE SOLVENTS, LOW VAPOR PRESSURE

The General Services Administration has authorized
the use of this federal specification by all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1. Scope. This specification covers the requirements for two types and two classes of organic wipe solvents used to prepare and clean aircraft and component surfaces prior to painting, sealing, and non-structural adhesive bonding. These solvents possess low vapor pressures and leave behind very low amounts of non-volatile residue upon evaporation.

1.2 Classification.

1.2.1 Type.

Type I - Fast evaporating
Flash point of 38-60°C (100-140°F).
Type II - Slow evaporating, high flash point
Minimum flash point of 61°C (140°F).

1.2.2 Class.

Class A - General use, not compatible with plastics.
Class B - Acrylic and polycarbonate compatible

Beneficial comments, recommendations, additions, deletions, etc., and any data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100 B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7930

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-A-250/4	-	Aluminum Alloy 2024, Plate and Sheet
QQ-A-250/5	-	Aluminum Alloy Alclad 2024, Plate and Sheet
QQ-A-250/12	-	Aluminum Alloy 7075, Plate and Sheet
QQ-A-250/13	-	Aluminum Alloy Alclad 7075, Plate and Sheet
CCC-C-46	-	Cloth, Cleaning, Nonwoven Fabric

DEPARTMENT OF DEFENSE

MIL-T-5624	-	Turbine Fuel, Aviation Grades JP-4, JP-5, and JP-5/JP-8 ST
MIL-P-8184	-	Plastic Sheet, Acrylic Modified
MIL-P-23377	-	Primer Coating: Epoxy, High Solids
MIL-L-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0-156
MIL-C-38334	-	Corrosion Removing Compound, Prepaint, for Aircraft Aluminum Surfaces
MIL-H-83282	-	Hydraulic Fluid, Fire Resistance, Synthetic Hydrocarbon Base, Aircraft, Metric, NATO Code Number H-537
MIL-P-83310	-	Plastic Sheet, Polycarbonate, Transparent
MIL-P-85582	-	Primer Coatings: Epoxy, Waterborne

FEDERAL

FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing
FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

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DEPARTMENT OF DEFENSE

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-290	-	Packaging of Petroleum and Related Products

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

DEPARTMENT OF LABOR

29 CFR 1910.1200	-	Occupational Safety and Health Standards - Hazard Communications

DEPARTMENT OF TRANSPORTATION

49 CFR 171-178	-	Subchapter C - Hazardous Materials Regulations

(Application of copies of the Code of Federal Regulations (CFR) should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are Department of Defense (DoD) adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI-Z129.1	-	American National Standard for the Precautionary Labeling of Hazardous Industrial Chemicals

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D56	-	Flash Point by Tag Closed Tester., Standard Test Method for
ASTM-D891	-	Specific Gravity of Liquid Industrial Chemicals, Standard Test Methods for.
ASTM-D1296	-	Odor of Volatile Solvents and Diluents, Standard Test Method for.
ASTM-D1353	-	Nonvolatile Matter in Volatile Solvents for Use in Paint, Varnish, Lacquer, and Related Products, Standard Test Method for.
ASTM-D1364	-	Water in Volatile Solvents (Fischer Reagent Titration Method), Standard Test Method for.
ASTM-D1613	-	Acidity in Volatile Solvents and Chemical Intermediates Used in Paint, Varnish, Lacquer, and Related Products, Standard Test Method for.
ASTM-D1876	-	Peel Resistance of Adhesives (T-Peel Test), Standard Test Method for.
ASTM-D1901	-	Relative Evaporation Time of Halogenated Organic Solvents and Their Admixtures., Standard Test Method for.
ASTM-D2879	-	Vapor Pressure-Temperature Relationship and Initial Decomposition Temperature of Liquids by Isoteniscope, Standard Test Method for.
ASTM-F22	-	Hydrophobic Surface Films by the Water-Break Test, Standard Test Method for.
ASTM-F483	-	Total Immersion Corrosion Test for Tank-Type Aircraft Maintenance Chemicals, Standard Test Method for.
ASTM-F484	-	Stress Crazing of Acrylic Plastics in Contact with Liquid or Semi-Liquid Compounds, Standard Test Method For.
ASTM-F945	-	Stress-Corrosion of Titanium Alloys by Aircraft Engine Cleaning Materials, Standard Test Method for.
ASTM-F1110	-	Sandwich Corrosion Test, Standard Test Method for.

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

SOCIETY FOR AUTOMOTIVE ENGINEERS (SAE)
AEROSPACE MATERIAL SPECIFICATIONS (AMS)

SAE-AMS-3276	-	Sealing Compound, Integral Fuel Tanks and General Purpose, Intermittent Use to 360°F (182°C).
SAE-AMS-3819	-	Cloths, Cleaning, For Aircraft Primary and Secondary Structural Surfaces.

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SAE-AMS-4377	-	Sheet and Plate, Magnesium Alloy, 3.0Al - 1.0Zn - 0.20Mn (AZ31B-H24), Cold Rolled, Partially Annealed.
SAE-AMS-4911	-	Sheet, Strip, and Plate, 6Al 4V, Annealed.
SAE-AMS-5040	-	Steel, Sheet and Strip, 0.15 Carbon, Maximum, Deep Forming Grade.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), samples shall be subjected to first article inspection in accordance with 4.3.

3.2 Toxicity. The wipe solvents supplied under this specification shall have no adverse effect on the health of personnel when used for its intended purpose. A Material Safety Data Sheet (MSDS) shall be prepared and submitted in accordance with FED-STD-313; additionally, it must conform to 29 CFR 1910.1200 (see 6.4).

3.3 Material. The wipe solvents supplied under this specification shall be clear, stable, and homogenous liquids, free of haze, rust particles, and suspended matter.

3.4 Prohibited materials. The wipe solvents shall not contain any substance listed in Appendix A or Appendix B.

3.5 Stability (Shelf-life). The wipe solvent shall be capable of being stored in its original, unopened container for a minimum of one year at a temperature of 34° ±1°C (90° ±2°F) without degrading, and shall be capable of meeting all the requirements of this specification after such storage.

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3.6 Performance requirements. The wipe solvent shall have the performance properties described in the table below. Failure an any test or inspection requirement specified herein shall resultin the rejection of that lot from which the sample came.

TABLE I. Performance requirements and test methods.

PROPERTY	REQUIREMENT	TEST PARAGRAPH	ASTM TEST METHOD
Nonvolatile content, maximum	200 parts per million (ppm) (0.02 gram/100 gram)	---	D1353
Sandwich corrosion, maximum	1	---	F1110
Immersion corrosion, maximum [milligram/square centimeter per day (mg/cm^2/day)]	---	---	---
Aluminum, Titanium, Steel, and Cadmium Plated Steel	0.01	4.5.1	F483
Magnesium	0.02	4.5.1	F483
Water, percent by weight, maximum	0.1	---	D1364
Acidity, as percent acetic acid, maximum	0.02	---	D1613
Evaporation rate, (Butylacetate = 1.0), minimum			
Type I	0.2	---	D1901
Type II	0.05	---	D1901
Compatibility with acrylic and polycarbonate material (Class B only)	No crazing, clouding, nor softening	4.5.2	F484
Flash point			
Type I	38° to 60°C (100° to 140°F)	---	D56
Type II	Minimum of 61°C (140°F)	---	D56
Odor	Not obnoxious	---	D1296
Specific gravity, 20°C/20°C	0.76 to 0.96	---	D891
Cleaning efficiency			
Lubricating oil (MIL-L-23699)	Water-break free	4.5.3	---
Hydraulic fluid (MIL-H-83282)	Water-break free	4.5.3	---
JP-5 jet fuel (MIL-T-5624)	Water-break free	4.5.3	---
Adhesion			
Wet tape test, primer	No primer peel	4.5.4.1	---
T-peel, sealant, minimum	10 pounds/inch (lb./in.)	4.5.4.2	---
Titanium stress corrosion, amount of cracking (examined with 500x magnification)	None	---	F945, method A
Vapor pressure, maximum (millimeters (mm) of Hg @ 20°C)	7	---	D2879

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use their own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. First article testing shall consist of all requirements and tests specified in section 3, and shall be accomplished on the first production lot of a contract or purchase order and when specified (see 6.2). Failure of any examination or test shall result in first article disapproval. The tests shall be performed by the manufacturer's own or any other facility suitable for the performance of this test, unless disapproved by the Government. Failure of any examination or test shall result in first article disapproval.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of all tests specified in section 3 and table I. Any failure shall constitute cause for rejection.

4.4.1 Lot formation. A lot shall consist of all material manufactured at one time from one batch, forming part of one contract, and submitted for acceptance. When required (see 6.2), the manufacturer shall furnish with each lot a certified test report showing that the material has passed the quality conformance inspection.

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4.5 Test methods. All tests performed for this specification shall be conducted in accordance with table I and 4.5.1 through 4.5.4.2.4. Unless otherwise stated in the test method or paragraph, room temperature is defined as 21° ±5°C (70° ±10°F) and relative humidity of 50 ±10 percent.

4.5.1 Immersion corrosion. This test shall be conducted in accordance with ASTM-F483 with test panels constructed of the following materials:

- Aluminum alloy 2024 (T3 temper), conforming to QQ-A-250/4;
- Aluminum alloy 7075 (T6 temper), conforming to QQ-A-250/12;
- Titanium (6AL-4V); conforming to SAE-AMS-4911
- Carbon steel (1010), conforming to SAE-AMS-5040
- Magnesium (AZ31B-H24), conforming to SAE-AMS-4377

4.5.2 Acrylic and polycarbonate plastic compatibility. This test shall be conducted in accordance with ASTM-F484 with type B test specimens conforming to MIL-P-8184, Type I or II, Class 2, (loaded at 24 megapascals (MPa) (3,500 pounds per square inch (psi)) for eight hours), and polycarbonate test specimens, machined from material conforming to MIL-P-83310, of the same dimensions as the acrylic test specimen and loaded to 14 MPa (2,000 psi) and tested for two hours.

4.5.3 Cleaning efficiency.

4.5.3.1 Test panel preparation. Prepare nine test panels, constructed from Alclad aluminum alloy 2024 (T3 temper) conforming to QQ-A-250/5, with approximate dimensions of 150.0 by 76.0 by 0.6 mm (6.0 by 3.0 by 0.025 inch (in.)). Deoxidize the test panels by immersion for two minutes in corrosion removing compound conforming to MIL-C-38334, Type I, Class 1. Upon removal, rinse the test panels with deionized water to obtain a water-break free surface. Allow test panels to air dry.

4.5.3.2 Test panel soiling. Three panels shall be individually soiled with each of the following contaminants:

a. Turbine engine lubricating oil conforming to MIL-L-23699
b. Hydraulic fluid conforming to MIL-H-83282
c. JP-5 jet fuel conforming to MIL-T-5624.

Soil the test panels by evenly spreading 5 ml of each contaminant onto the surface of the test panel to be cleaned and coated with primer coating. Store the lubricating oil and hydraulic fluid

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contaminated test panels for 24 hours at 49° ±1°C (120° ±2°F). Air dry the jet fuel contaminated test panels 24 hours at room temperature.

4.5.3.3 Procedure. Clean the nine test panels as follows:

a. Fold a wipe cloth, conforming to CCC-C-46, Type I, Class 7, or to AMS 3819, Grade A, and initial approximate dimensions of 228 by 228 mm (9 by 9 in.), into a square of about 114 mm by 114 mm (4.5 by 4.5in.).
b. Soak the folded cloth with no more than 10 milliliters (ml) of test solvent.
c. Wipe each test panels with one stroke in one direction, applying moderate manual pressure. Ensure that the contact area between the cloth and the test panel is saturated with the test solvent. This procedure (a. through c.) may be repeated no more than three times.
d. Allow to air dry for no more than one hour at ambient conditions (see 4.5).
e. Examine each test panel for conformance to table I, in accordance with ASTM-F22. A water-break free surface is considered passing.

4.5.4 Adhesion.

4.5.4.1 Wet tape.

4.5.4.1.1 Test panel preparation. Prepare eighteen test panels, constructed from Alclad aluminum alloy 2024 (T3 temper) conforming to QQ-A-250/5, with approximate dimensions of 150.0 by 76.0 by 0.6 mm (6.0 by 3.0 by 0.025 in.). Deoxidize the test panels by immersion for two minutes in corrosion removing compound conforming to MIL-C-38334, Type I, Class 1. Upon removal, rinse with deionized water to obtain a water-break free surface. Allow test panels to air dry.

4.5.4.1.2 Test panel soiling. Six test panels shall be individually soiled with each of the following contaminants (Soil three test panels with each):

a. Turbine engine lubricating oil conforming to MIL-L-23699
b. Hydraulic fluid conforming to MIL-H-83282
c. JP-5 jet fuel conforming to MIL-T-5624.

Soil the test panels by evenly spreading five milliliters of contaminant onto the test surface of the test panel. Store the lubricating oil and hydraulic fluid contaminated test panels for 24 hours at 49° ±1°C (120° ±2°F). Air dry the jet fuel contaminated test panels for 24 hours at room temperature.

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4.5.4.1.3 Test panel cleaning. Lightly dampen a wipe cloth conforming to AMS 3819, Grade A, or CCC-C-46, Class 7, with the test solvent and clean the soiled test panels with one stroke in one direction, applying moderate manual pressure. This procedure may be repeated no more than three times, using a clean wipe cloth and changing stroke direction each time. Allow the test panels to air dry.

4.5.4.1.4 Application of primer coating. Spray apply primer coating conforming to MIL-P-23377, Type I, Class C, to three soiled test panels of each contaminant within one hour of cleaning with the test solvent. Spray apply primer coating conforming to MIL-P-85582, Type I, Class C1 or N, to the remaining test panels within one hour of cleaning with the test solvent. Store the coated test panels at 22° ±2°C (72° ±4°F) for a minimum of 14 days.

4.5.4.1.5 Procedure. Test the coated test panels in distilled water for 24 hours at room temperature, in accordance with FED-STD-141, Method 6301. Examine the primer coatings for conformance to Table I There shall be no damage to the surface adhesion of any of the primer coatings.

4.5.4.2 T-peel.

4.5.4.2.1 Test panel preparation. Prepare twelve test panels, constructed from Alclad aluminum alloy 2024 (T3 temper), conforming to QQ-A-250/5, with approximate dimensions of 304.8 by 152.4 by 0.41 mm (12.0 by 6.0 by 0.016 in). Deoxidize the test panels by immersing each for two minutes in corrosion removing compound conforming to MIL-C-38334, Type I, Class 1. Upon removal, rinse with deionized water to obtain a water-break free surface. Allow test panels to air dry.

4.5.4.2.2 Test panel soiling. Soil twelve test panels, four test panels for each contaminant:

a. Hydraulic fluid conforming to MIL-H-83282
b. Turbine engine lubricating oil conforming to MIL-L-23699
c. JP-5 jet fuel conforming to MIL-T-5624.

Soil the test panels by evenly spreading 5 ml of contaminant onto the test surface of each test panel. Store the lubricating oil contaminated and hydraulic fluid contaminated test panels for 24 hours at 49° ±1°C (120° ±2°F). Air dry the jet fuel contaminated test panels for 24 hours at room temperature (see 4.5).

4.5.4.2.3 Test panel cleaning. Lightly dampen a wipe cloth conforming to AMS 3819, Grade A, or CCC-C-46, Class 7, with the test solvent and clean the test panels with one stroke in one direction, applying moderate manual pressure. This procedure may be repeated no more than

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three times, using a clean wipe cloth each time, and changing stroke direction each time. Allow the test panels to air dry.

4.5.4.2.4 Procedure.

a. On six pairs of panels, two pairs for each contaminant, mask the upper 68.0 mm (3.0 in.) of each test panel by applying Teflon® masking tape.
b. Apply sealant conforming to AMS 3276, Class A-2, to all of the test panels over the lower 230 mm (9 in.) of each test panels. Do not apply sealant to the upper 68.0 mm (3.0 in.) of each test panel.
c. After application of the sealant, place one layer of 0.13 mm (0.005 in) thick scrim cloth on top of the sealant on one of the test panels (the scrim cloth will cover 3/4 of the surface panel). Do not cover the Teflon® taped portion of the test panel.
d. On top of the first panel, place another test panel, which has been soiled with the same contaminant, cleaned, and had sealant applied, ensuring that the Teflon® taped portions of the two panels are initially in contact prior to joining the sealant covered portions of the panels.
e. With a roller weighing 1.8 kilogram (kg) (4 lb), and starting at the Teflon®-taped end of the panel, apply manual pressure and push the roller over the surface of the joined panels. Maintain a uniform bond line thickness of the scrim cloth.
f. Wipe off the excess sealant that is squeezed out.
g. While the sealant between the test panels is curing, maintain a minimum uniform pressure of 34 kilopascals (kPa) (5 psi) on the test panels to control thickness and promote uniform contact between the bonding surfaces.
h. Cure the test panels for a minimum of 14 days at 24° ±2°C (75° ±4°F) and 50 percent relative humidity.
i. Cut the two sets of bonded test panels into strips measuring 304.8 by 25.4 by 0.41 mm (12.0 by 1.0 by 0.016 in.). To eliminate edge effects, cut the first strip 12.7 mm (0.5 in.) from the edge of the 152.4 mm (6.0 in.) wide test panels. A total of five 25.4 mm (1.0 in.) wide strips and two 12.7 mm (0.5 in.) wide strips will result. In accordance with ASTM-D1876, test the 25.4 mm (1.0 in.) test strips of each contaminant for conformance to table I.

5. PACKAGING

5.1 Packaging. The packaging shall conform to MIL-STD-290, level Commercial, unless otherwise specified by the procurer (see 6.2).

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5.2 Marking. All marking shall conform to MIL-STD-129 and MIL-STD-290. In addition, individual containers shall bear printed labels with the following:

- Federal Specification P-W-2891, Wipe Solvents, Low Vapor Pressure
- Type and Class
- Manufacturer's name and product number.
- Date of manufacture by month and year.
- Batch number.
- Net contents.
- Flashpoint.

All unit and intermediate packs of toxic and hazardous chemicals and materials shall be labeled in accordance with all applicable laws, statutes, regulations and ordinances, including federal, state, and municipal requirements. All unit containers (including those that serve as shipping containers) shall be marked in accordance with 49 CFR 171-178 and with the applicable precautionary information detailed in ANSI Z129.1.

5.2.1 Type I. Type I wipe solvents shall bear the following special marking:

"WARNING: This material is not permitted to be used on-board United States Navy ships"

6. INFORMATION FOR GUIDANCE ONLY

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This material supplied under this specification is intended to be used for hand-wipe cleaning of most aluminum, titanium, steel, and magnesium aircraft, and other aircraft component surfaces prior to painting, sealing, and non-structural adhesive bonding. These hand-wipe solvents are not intended to be sprayed, misted, or atomized in any manner; the solvents are intended to be poured or squirted onto wipe cloths for hand-wipe cleaning applications. The Type I wipe solvent is not approved for use on board United States Navy ships.

6.2 Acquisition requirements. Acquisition documents should specify the following, as a minimum:

a. Title, number and date of this specification, including any amendments.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type and class of wipe solvent required (see 1.2).
d. Quantity and size of containers (see 6.5).

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e. If a certified test report is required for the quality conformance inspection (see 4.4.1)
f. Packaging level, if other than Commercial packaging is required.
g. If First Article Inspection is required (see 3.1 and 4.3). If First Article inspection is required, specify the laboratory to perform inspection.
h. Address to whom MSDSs should be sent.

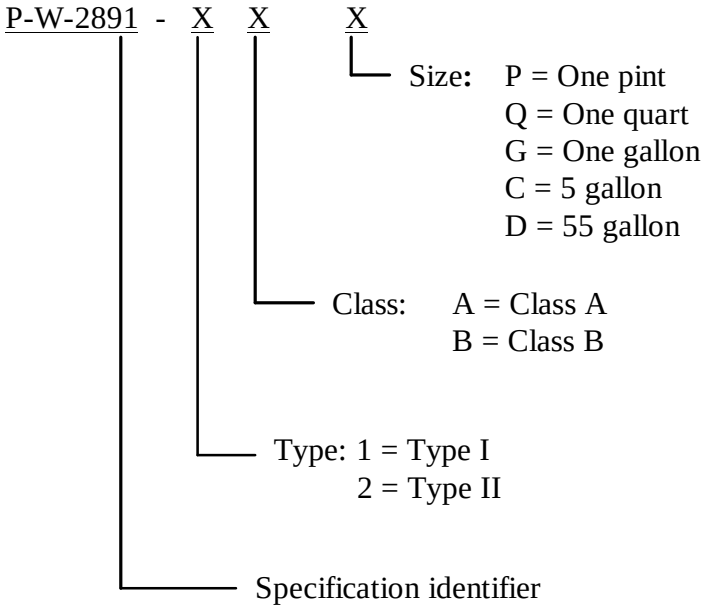
6.3 Subject term (key word) listing.

Non-volatile residue
Prepaint
Preseal
Solvent
Wipe cleaner

6.4 Material Safety Data Sheet (MSDS). 29 CFR 1910.1200 requires that the MSDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the MSDS.

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6.5 Part or identifying number (PIN). The following part numbering procedure is for government purposes and does not constitute a requirement:



Custodians:
Navy - AS

Civil coordinating agency:
GSA - FSS

Preparing activity:
Navy - AS

(DoD Project 7930-0424)

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APPENDIX A

CLASS 1 OZONE DEPLETING SUBSTANCES (ODSs)

A.1 SCOPE

A.1.1 Scope. This appendix provides a listing of class 1 ODSs, as defined by the Clean Air Act Amendment of 1990, and its subsequent Amendments, including their Chemical Abstract Service (CAS) Registry name and number.

Common Name	CAS Name	CAS Number
Halon 1211	Bromochlorodifluoromethane	353-59-3
Halon 1301	Bromotrifluoromethane	75-63-8
Halon 2402	Dibromotetrafluoroethane	124-73-2
CFC-11	Trichlorofluoromethane	75-69-4
CFC-12	Dichlorodifluoromethane	75-71-8
CFC-13	Chlorotrifluoromethane	75-72-9
CFC-111	Pentachlorofluoroethane	954-56-3
CFC-112A	Tetrachlorodifluoroethane	76-11-9
CFC-112	Tetrachlorodifluoroethane	76-12-0
CFC-113	Trichlorotrifluoroethane	76-13-1
CFC-114	Dichlorotetrafluoroethane	76-14-2
CFC-115	Chloropentafluoroethane	76-15-3
CFC-211	Heptachlorofluoropropane	422-78-6
CFC-212	Hexachlorodifluoropropane	3182-26-1
CFC-213	Pentachlorotrifluoropropane	2354-06-5
CFC-214	Tetrachlorotetrafluoropropane	29255-31-0
CFC-215	Trichloropentafluoropropane	4259-43-2
CFC-216	Dichlorohexafluorpropane	661-97-2
CFC-217	Chloroheptafluoropropane	422-86-6
Methyl Chloroform	1,1,1-Trichloroethane	71-55-6
Tetrachloromethane	Carbon Tetrachloride	56-23-5

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APPENDIX B

HAZARDOUS AIR POLLUTANTS (HAPs) SUBJECT TO THE PROVISIONS OF SECTION 112 OF THE 1990 CLEAN AIR ACT AMENDMENTS

B.1 SCOPE

B.1.1 Scope. This appendix provides a cross reference of Chemical Abstract Service (CAS) Registry number and its associated chemical name.

CAS Number	Chemical Name
75-07-0	Acetaldehyde
60-35-5	Acetamide
75-05-8	Acetonitrile
98-86-2	Acetophenone
53-96-3	2-Acetylaminofluorene
107-02-8	Acrolein
79-06-1	Acrylamide
79-10-7	Acrylic acid
107-13-1	Acrylonitrile
107-05-1	Allyl chloride
92-67-1	4-Aminobiphenyl
62-53-3	Aniline
90-04-0	o-Anisidine
1332-21-4	Asbestos
71-43-2	Benzene(including benzene from gasoline)
92-87-5	Benzidine
98-07-7	Benzotrichloride
100-44-7	Benzyl chloride
92-52-4	Biphenyl
117-81-7	Bis(2-ethylhexyl)phthalate (DEHP)
542-88-1	Bis(chloromethyl)ether
75-25-2	Bromoform
106-99-0	1,3-Butadiene
156-62-7	Calcium cyanamide
105-60-2	Caprolactam
133-06-2	Captan
63-25-2	Carbaryl
75-15-0	Carbon disulfide
56-23-5	Carbon tetrachloride

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CAS Number	Chemical Name
463-58-1	Carbonyl sulfide
120-80-9	Catechol
133-90-4	Chloramben
57-74-9	Chlordane
7782-50-5	Chlorine
79-11-8	Chloroacetic acid
532-27-4	2-Chloroacetophenone
108-90-7	Chlorobenzene
510-15-6	Chlorobenzilate
67-66-3	Chloroform
107-30-2	Chloromethyl methyl ether
126-99-8	Chloroprene (Neoprene; 2-Chloro-1,3-butadiene)
1319-77-3	Cresols/Cresylic acid (isomers and mixtures)
108-39-4	m-Cresol
95-48-7	o-Cresol
106-44-5	p-Cresol
98-82-8	Cumene
94-75-7	2,4-D, salts and esters
3547-04-4	DDE
334-88-3	Diazomethane
132-64-9	Dibenzofuran
96-12-8	1,2-Dibromo-3-chloropropane
84-74-2	Dibutylphthalate
106-46-7	1,4-Dichlorobenzene(p)
91-94-1	3,3-Dichlorobenzidene
111-44-4	Dichloroethyl ether (Bis(2-chloroethyl)ether)
542-75-6	1,3-Dichloropropene
62-73-7	Dichlorvos
111-42-2	Diethanolamine
121-69-7	N,N-Dimethylaniline
64-67-5	Diethyl sulfate
119-90-4	3,3-Dimethoxybenzidine
60-11-7	Dimethyl aminoazobenzene
119-93-7	3,3-Dimethyl benzidine
79-44-7	Dimethyl carbamoyl chloride
68-12-2	Dimethyl formamide
57-14-7	1,1-Dimethyl hydrazine

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CAS Number	Chemical Name
131-11-3	Dimethyl phthalate
77-78-1	Dimethyl sulfate
534-52-1	4,6-Dinitro-o-cresol
51-28-5	2,4-Dinitrophenol
121-14-2	2,4-Dinitrotoluene
123-91-1	1,4-Dioxane (1,4-Diethyleneoxide)
122-66-7	1,2-Diphenylhydrazine
106-89-8	Epichlorohydrin (1-Chloro-2,3-epoxypropane)
106-88-7	1,2-Epoxybutane (1,2-Butylene oxide)
140-88-5	Ethyl acrylate
100-41-4	Ethyl benzene
51-79-6	Ethyl carbamate (Urethane)
75-00-3	Ethyl chloride (Chloroethane)
106-93-4	Ethylene dibromide (1,2-Dibromoethane)
107-06-2	Ethylene dichloride (1,2-Dichloroethane)
107-21-1	Ethylene glycol
151-56-4	Ethylene imine (Aziridine)
75-21-8	Ethylene oxide
96-45-7	Ethylene thiourea
75-34-3	Ethylidene dichloride (1,1-Dichloroethane)
50-00-0	Formaldehyde
76-44-8	Heptachlor
118-74-1	Hexachlorobenzene
87-68-3	Hexachlorobutadiene
77-47-4	Hexachlorocyclopentadiene
67-72-1	Hexachloroethane
822-06-0	Hexamethylene-1,6-diisocyanate
680-31-9	Hexamethylphosphoramide
110-54-3	Hexane
302-01-2	Hydrazine
7647-01-0	Hydrochloric acid
7664-39-3	Hydrogen fluoride (Hydrofluoric acid)
7783-06-4	Hydrogen sulfide
123-31-9	Hydroquinone
78-59-1	Isophorone
58-89-9	Lindane (all isomers)
108-31-6	Maleic anhydride
67-56-1	Methanol

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CAS Number	Chemical Name
72-43-5	Methoxychlor
74-83-9	Methyl bromide (Bromomethane)
74-87-3	Methyl chloride (Chloromethane)
71-55-6	Methyl chloroform (1,1,1-Trichloroethane)
78-93-3	Methyl ethyl ketone (2-Butanone)
60-34-4	Methyl hydrazine
74-88-4	Methyl iodide (Iodomethane)
108-10-1	Methyl isobutyl ketone (Hexone)
624-83-9	Methyl isocyanate
80-62-6	Methyl methacrylate
1634-04-4	Methyl tert butyl ether
101-14-4	4,4-Methylene bis (2-chloroaniline)
75-09-2	Methylene chloride (Dichloromethane)
101-68-8	Methylene diphenyl diisocyanate (MDI)
101-77-9	4,4-Methylenedianiline
91-20-3	Naphthalene
98-95-3	Nitrobenzene
92-93-3	4-Nitrobiphenyl
100-02-7	4-Nitrophenol
79-46-9	2-Nitropropane
684-93-5	N-Nitroso-N-methylurea
62-75-9	N-Nitrosodimethylamine
59-89-2	N-Nitrosomorpholine
56-38-2	Parathion
82-68-8	Pentachloronitrobenzene (Quintobenzene)
87-86-5	Pentachlorophenol
108-95-2	Phenol
106-50-3	p-Phenylenediamine
75-44-5	Phosgene
780-35-12	Phosphine
7723-14-0	Phosphorus
85-44-9	Phthalic anhydride
1336-36-3	Polychlorinated biphenyl (Aroclors)
1120-71-4	1,3-Propane sultone
57-57-8	beta-Propiolactone
123-38-6	Propionaldehyde
114-26-1	Propoxur (Baygon)
78-87-5	Propylene dichloride (1,2-Dichloropropane)

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CAS Number	Chemical Name
75-56-9	Propylene oxide
75-55-8	1,2-Propylenimine (2-Methyl aziridine)
91-22-5	Quinoline
106-51-4	Quinone (1,4-Cyclohexadienedione)
100-42-5	Styrene
96-09-3	Styrene oxide
1746-01-6	2,3,7,8-Tetrachlorodibenzo-p-dioxin
79-34-5	1,1,2,2-Tetrachloroethane
127-18-4	Tetrachloroethylene (Perchloroethylene)
7550-45-0	Titanium tetrachloride
108-88-3	Toluene
95-80-7	2,4-Toluene diamine
584-84-9	2,4-Toluene diisocyanate
95-53-4	o-Toluidine
8001-35-2	Toxaphene (chlorinated camphene)
120-82-1	1,2,4-Trichlorobenzene
79-00-5	1,1,2-Trichloroethane
79-01-6	Trichloroethylene
95-95-4	2,4,5-Trichlorophenol
88-06-2	2,4,6-Trichlorophenol
121-44-8	Triethylamine
1582-09-8	Trifluralin
540-84-1	2,2,4-trimethylpentane
108-05-4	Vinyl acetate
593-60-2	Vinyl bromide
75-01-4	Vinyl chloride
75-35-4	Vinylidene chloride (1,1-Dichloroethylene)
1330-20-7	Xylenes (isomers and mixture)
108-38-3	m-Xylene
95-47-6	o-Xylene
106-42-3	p-Xylene
	Antimony compounds
	Arsenic compounds (inorganic including arsine)
	Beryllium compounds
	Cadmium compounds
	Chromium compounds
	Cobalt compounds

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CAS Number	Chemical Name
	Coke oven emissions
	Cyanide compounds 1/
	Glycol ethers 2/
	Lead compounds
	Manganese compounds
	Mercury compounds
	Fine Mineral fibers 3/
	Nickel compounds
	Polycyclic organic matter 4/
	Radionuclides (including radon) 5/
	Selenium compounds

NOTE: For all listings above which contain the word "compounds" and for glycol ethers, the following applies: Unless otherwise specified, these listings are defined as to include any unique chemical substance that contains the named chemical (antimony, arsenic, etc.) as part of that chemical's infrastructure.

1/ X' CN where X = H' or any other group where a formal dissociation may occur. For example KCN or $Ca(CN)_2$.

2/ Includes mono- and di- ethers of ethylene glycol, diethylene glycol, and triethylene glycol

$R\text{-}(OCH_2CH_2)_n\text{-}OR'_1$

where n = 1,2, or 3

R = alkyl or aryl groups

R' = R, H, or groups which, when removed, yield glycol ethers with the structure: $R\text{-}(OCH_2CH)_n\text{-}OH$. Polymers are excluded from the glycol category.

3/ Includes mineral fiber emissions from facilities manufacturing or processing glass, rock, or slag fibers (or other mineral derived fibers) of average diameter 1 micrometer or less.

4/ Includes organic compounds with more than one benzene ring, and which have a boiling point greater than or equal to 100°C.

5/ A type of atom which spontaneously undergoes radioactive decay.

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